Solar Array Deployement

Final Report for the collaborative project of the Multibody System Dynamics course

Group number 9-E

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1. Problem statement

The aim of the work is to reproduce and study the behavior of the deployment mechanism of the solar panels on the Lunar Reconnaissance Orbiter. The system of interest consists of three panels initially found in closed configuration, attached to the body of the satellite as depicted in Figure 1.

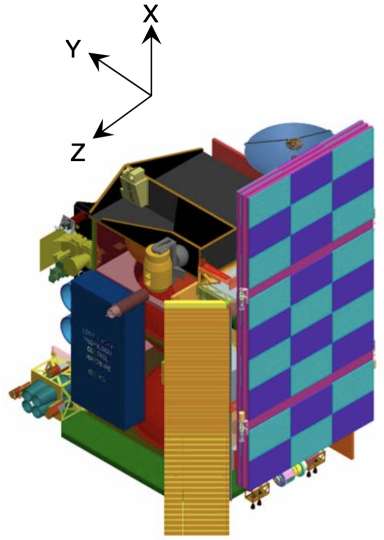

Figure 1: Closed configuration

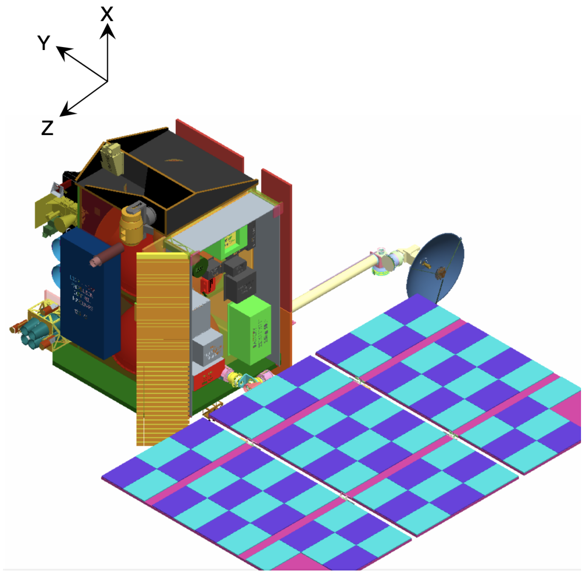

Figure 2: Open configuration

The driving mechanism consists of a DC motor, which allows the rotation of the solar array assembly, and a system of four precharged springs which, together with two dampers, regulate the side panels' opening. The sequence is shown in the following Figure 3 and Figure 4. During the first act, the closed assembly reaches the 90° of aperture in 300 seconds with respect to the body of the satellite. Ended the first motion, the left panel (seen from the satellite) starts opening thanks to the activation of the prestrained joints, through which it is linked to the central panel. Once it reaches 60° in the angular motion, the right one starts moving. The system deployment is stopped when both the side panels rotated of 180° with respect to their original position.

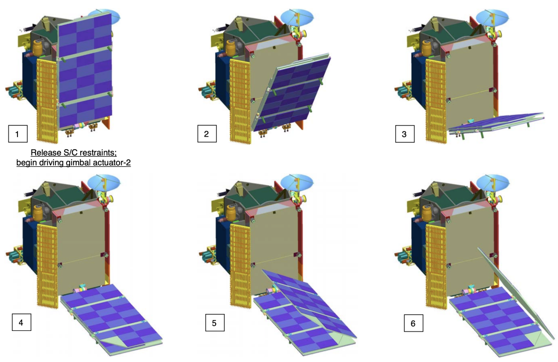

Figure 3: Deployment sequence

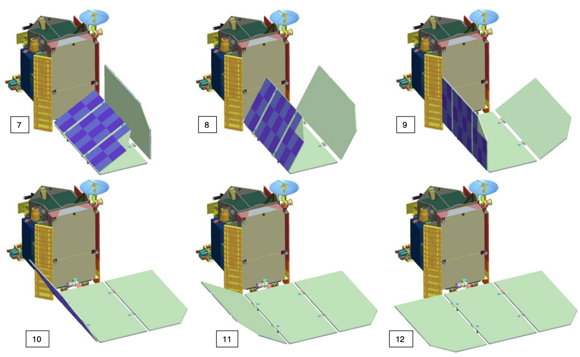

Figure 4: Deployment sequence

Three different models were therefore developed: a kinematic one on Adams, used to retrieve values of the needed couples to correctly open the panels and to size the spring-damper system; a dynamic model on MBDyn which used the found data, to retrieve positions and velocities of the points of interests and a Matlab one implementing a simplified version of the system to individuate discrepancies with the other two models and to develop the control law of the motor.

2. ADAMS Model

2.1. CAD model

In order to examine the deployment sequence in Adams, first the CAD model was needed. Based on the data provided in the LRO preliminary design review [1] panels and hinges were modelled in Siemens NX. The goal was to have a realistic model that would accurately represent the satellite. Hinges' dimensions were not provided in the document so their properties were chosen based on other geometrical values. The final model is presented below.

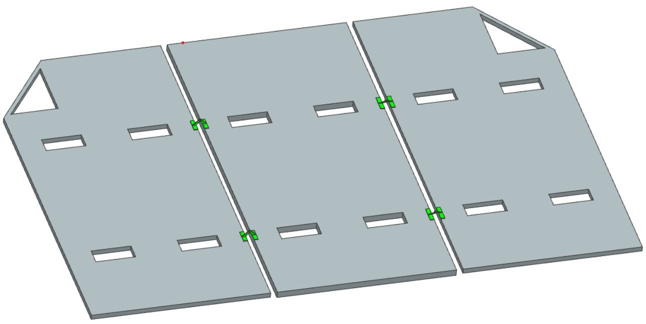

Figure 5: Panels open

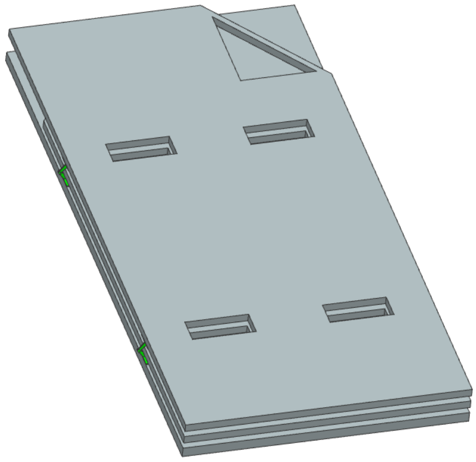

Figure 6: Panels closed

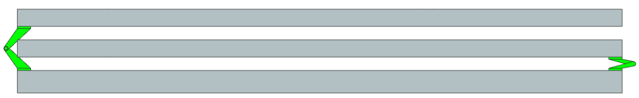

Figure 7: View from the bottom

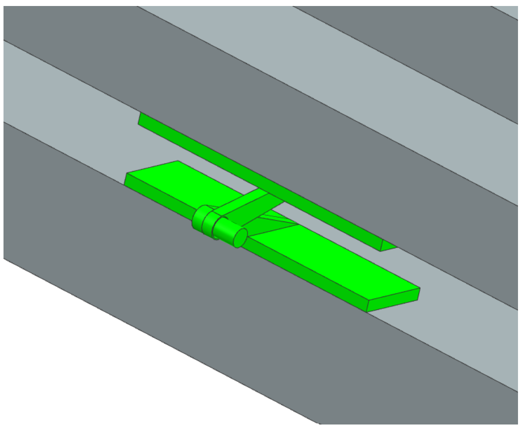

Figure 8: Hinge closeup

2.2. Basic Adams setting

Next the model in a *STEP* format was imported to Adams. Based on the volume properties of elements and masses of each panel provided in [1] the average densities were inputted.
Later on, the joints were made to connect all the elements. Firstly, the middle panel was set

as ground. Next, hinges were connected to panels by fixed joints. Later, hinges were connected by spherical joints. This decision was taken because in the end the result was identical to using double cylindrical joints, however turned out to be easier to implement it in Adams. The resultant multibody system had two degrees of freedom.
To one hinge in each pair, a spring and a damper were added. After in depth considerations and analysing the results of the simulations, the stiffness and damping coefficients were sized in order to guarantee convergence in less then 100 seconds while maintaining an angular velocity lower than 10 deg/s. Results of the sizing are shown in Table 3.
The holding mechanism was modelled as a contact force acting between panels (left panel-right panel; right panel- middle panel). For the assembly to open, a step function was designed to set the contact forces from holding to 0N in 0.1N. Each panel maximum opening angle was set to 180° and made by introducing another contact force (left panel-middle panel; right panel-middle panel).

2.3. Further Adams model development

To model the servo controlling the solar array opening sequence the following changes were made:

1. The schematic body of a satellite was designed in Adams and set as a new ground.
2. New joint mechanism between middle panel and satellite body were designed. These consisted of two independent cylindrical joints allowing the movement of the panels as it happens in reality.

The opening sequence of the solar array was implemented with the use of motion function acting on one of the servo joints. The chosen solution was a step5 function, since it guarantees very low maximum torque, described below:

$$STEP5(x, x_0, h_0, x_1, h_1)$$

x- independent variable - time
x_0- initial starting point- time= 0s
x_1- ending point- time= 300s
h_0- starting value- angle= 0°
h_1- final value- angle= 90°

The end motion results in the solar array opening from 0° to 90° in 300 s. The motion function presents a continuous first and second derivatives.
At the end the additional motion of the fully deployed panel array was added to simulate the system adjustment its position to the sun. The procedure was the same as described above.
The full opening and operating procedure simulated is described below:

1. Time= 0s- Beginning of the opening of the array
2. Time= 300s- Array reaches 90°
3. Time= 300s- The left panel holding force (locking mechanism) goes to 0N in 0.1s
4. Left panel reaches 60°
5. The right panel holding force (locking mechanism) goes to 0N in 0.1s
6. First left then right panel reach 180−° opening angle
   (Here the further animation part of simulation is described)
7. Time= 700s Panel array open further by 45° using step5 function
8. Second servo joint turns the panel around animating the sun tracking procedure

Figure 9 shows the motion imposed to the motor joint; Figure 10 shows the angular position and velocity obtained with the sized spring-damper systems.

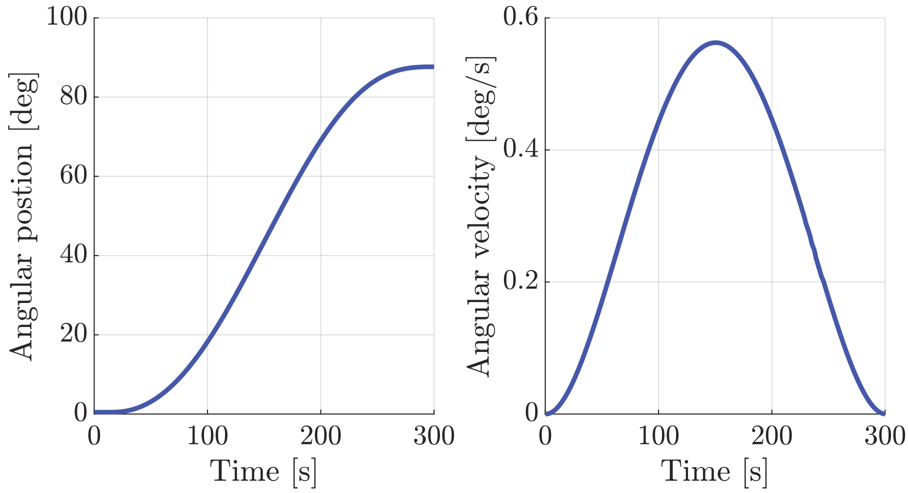

Figure 9: First Deployment

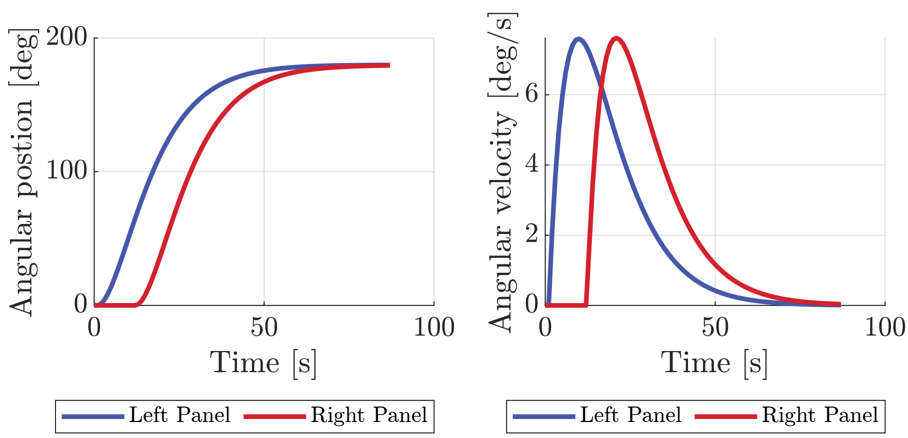

Figure 10: Second Deployment

Through the above explained and shown motion imposition it was possible to retrieve the intensity of the torque needed to correctly execute the motion. The moment curve is displayed in Figure 11.

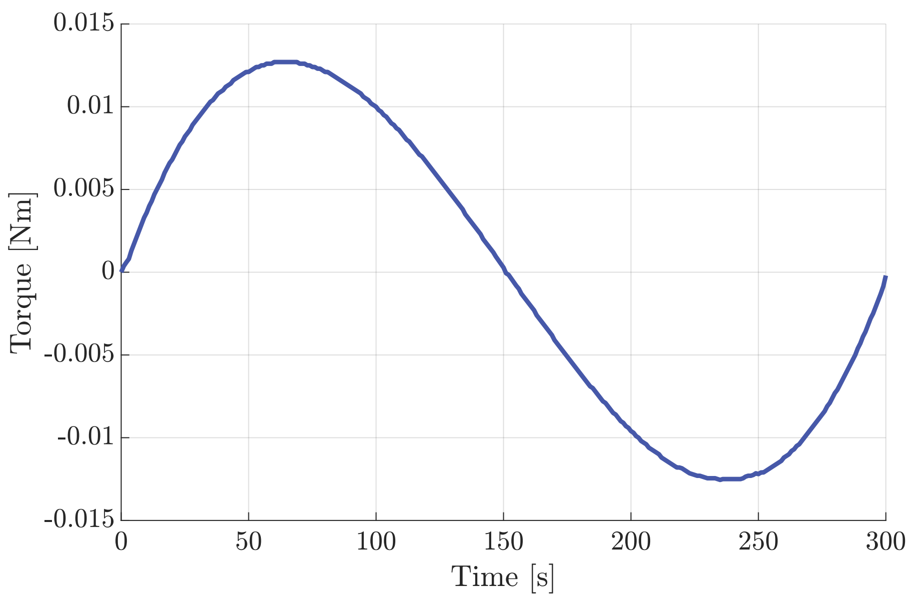

Figure 11: Torque

2.4. Simulation animation

The images below show some pictures from the animation sequence fully available at the following link.

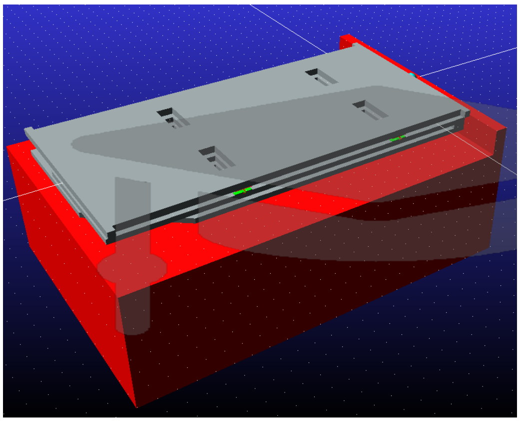

Figure 12: Closed satellite

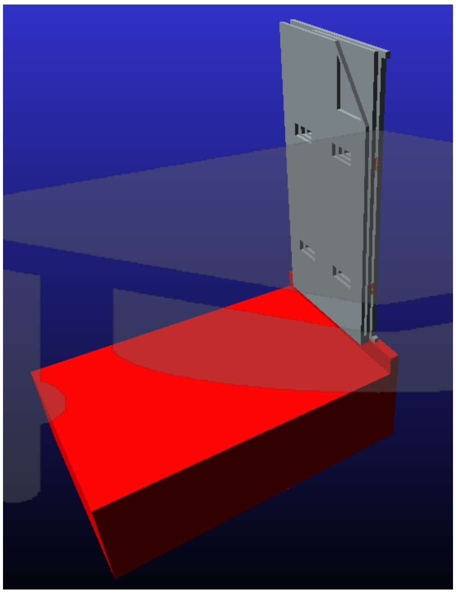

Figure 13: Sollar assemlby rotated by 90°

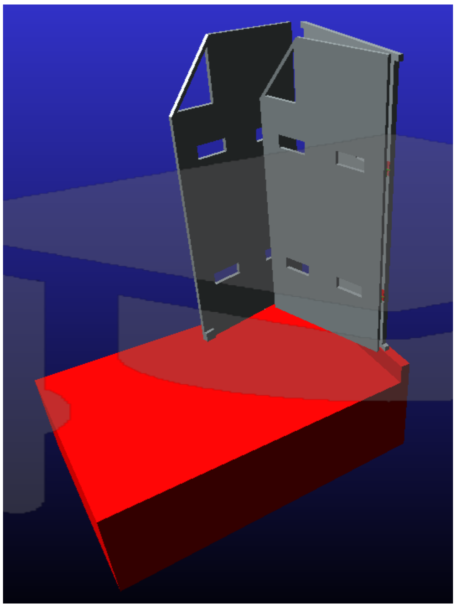

Figure 14: Opening of panels

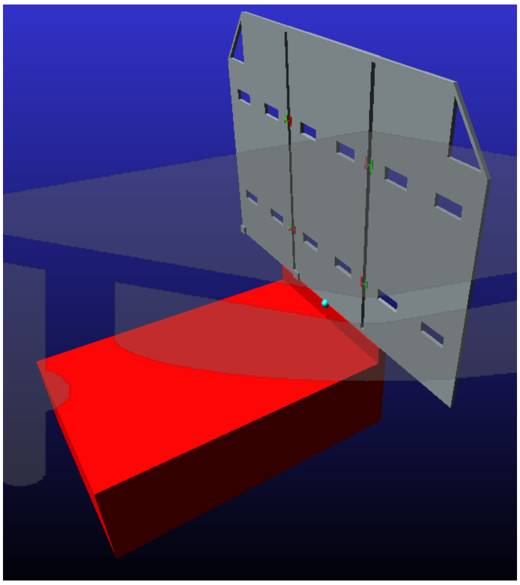

Figure 15: Panels fully opened

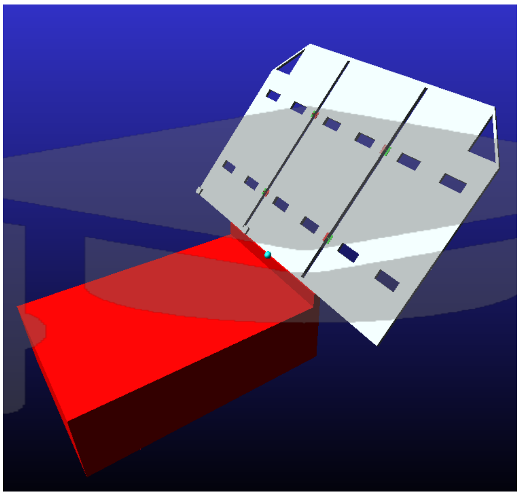

Figure 16: Sun tracking

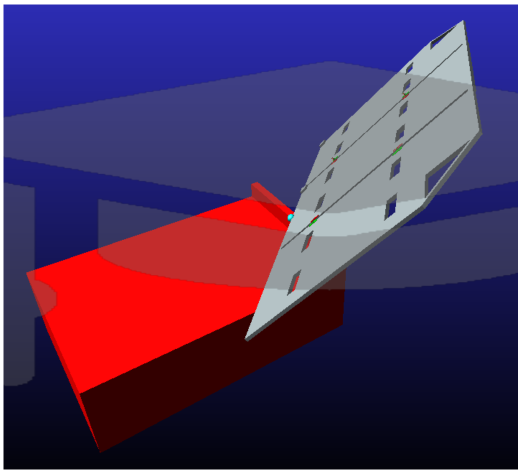

Figure 17: Sun tracking

2.5. Conclusion

It has turned out that modeling of the 3D multibody systems in Adams software is subjectively complicated and relatively time consuming. It is wise to fully develop the model in CAD software first, since any alterations of physical model are highly complicated in Adams. Even small detail like taking into consideration contact forces at the beginning of the process is worth thinking through. More often than not choosing appropriate markers for measurements of the model properties was not a trivial task. The process was however, very educational. Many problems had to be addressed: modeling joints, imposing motion functions to choosing realistic spring properties, just to name at few.
The end model consisted of thirteen bodies and fourteen connectors. Ten forces were controlled and the system was left with four degrees of freedom.

3. MATLAB Model

The objectives of the MATLAB models are to create a simplified version of the system to analyze its dynamics response during the deployment and to design a controller for the DC motor used in the first part of the above mentioned process.

Each of the three panels has been modeled as a rigid beam with mass m, length l. Each body is considered as independent from the others and is constrained to the ground using a *revolute hinge* at one of its extremities. As a consequence, the beam is only able to rotate around an axis passing through the hinge and perpendicular to the beam plane, with an inertia $J = \frac{1}{3}ml^2$.

3.1. Side panels deployment

Each side panel is connected to the ground also via a torsional spring with stiffness coefficient K computed as the sum of the stiffnesses of each of the springs attaching it to the middle panel and with a damper of coefficient C. The panels starts from an angular position of 0° and the springs present a prestrain of 180°. The joints parameters are the one sized using Adams and are summarized in Table 3.
The overall equations governing the motion of the two side panels are the one of a typical

spring-damper system:

$$\mathrm{M}\ddot{\boldsymbol{\vartheta}} + \mathrm{C}\dot{\boldsymbol{\vartheta}} + \mathrm{K}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta_0}) = 0 \tag{1}$$

where $\boldsymbol{\vartheta}$ is the vector of the angular positions of the two panels, $\boldsymbol{\vartheta_0}$ is the vector of prestrain of the springs and M, C, K are the inertia, damping and stiffness matrices respectively.
In order to simulate the forces generated by the latches that prevents the panels from moving, two different strategies were implemented. Before the panels need to move, the prestrain of the corresponding spring is set to 0; once the panels reach the final position, their velocity and acceleration are simply set to 0.
Figure 18 shows the simulated positions and velocities of both side panels.

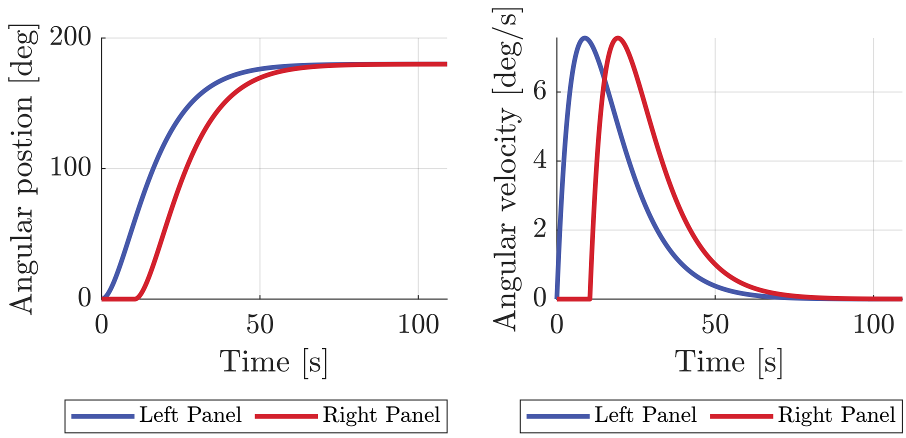

Figure 18: Angular position and velocity of the two side panels during deployment

3.2. DC motor model

As previously stated, the first part of the deployment requires an electric motor to rotate of 90 degrees the solar array assembly. Given the maximum torque of around 0.015 Nm found in the sizing process done in MSC Adams, the DB-1500 J-1ES brushless DC motor from MOOG, which guarantees a peak torque of 0.15 Nm, has been modeled [2]. The motor is mechanically characterized by a moment of inertia J_{motor} and is able to generate a torque proportional to the current I flowing in the armature coils, which are modeled as a resistance R and an inductance L in series. On the armature also acts a back electromotive force proportional to the angular velocity of the system.
The complete set of equations governing the motion of the coupled motor-solar array systems is represented in Equation 2.

$$\begin{cases} J_{sys} = J_{motor} + J_{array} \\ T = K_m I \\ \dot{\omega} = \dfrac{T}{J_{sys}} \\ \dot{\vartheta} = \omega \\ V = K_m\omega + RI + L\dot{I} \end{cases} \tag{2}$$

In order to effectively control the deployment, a simple cascade controller has been implemented.
A PI controller regulates the inner loop of the system, guaranteeing fast convergence of the

electric dynamics of the motor. It was designed neglecting the back EMF force and such that the open-loop system behaves as an integrator with an high cut-off frequency (500Hz).
The outer-loop controller is a PD controller designed assuming steady-state behaviour of the inner loop. It has a slow zero to guarantee stability and a very fast pole to ensure feasibility. The gain is designed such that the cut-off frequency of the open-loop system is 1Hz.

The block diagram of the overall system is shown in Figure 19.

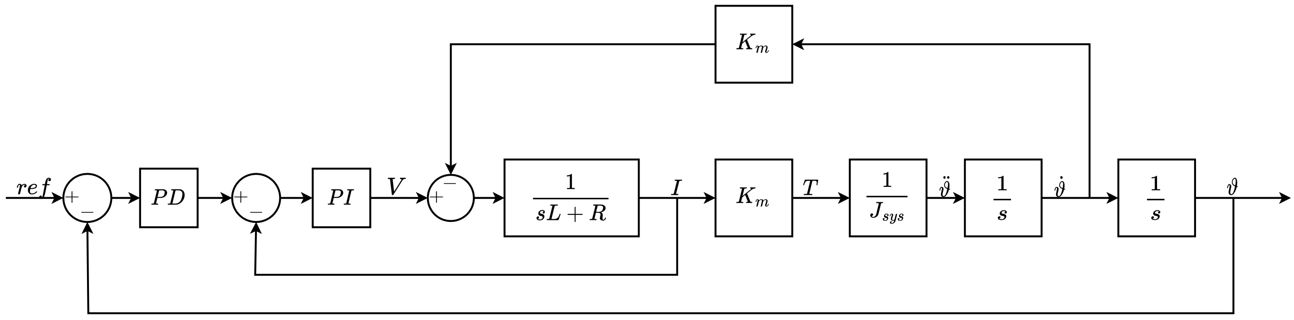

Figure 19: First deployment model's block diagram

As it has been done in MSC Adams, the reference position commanded to the system is a STEP5 function which goes from 0° to 90° in 300 seconds.

Figure 20 shows the evolution of the voltage and the current in the coils of the motor during the deployment. It can be seen that they are well below the rated limit of 4V and 15A respectively. Also the output torque is always much lower that the maximum 0.15Nm, as can be seen in Figure 21.
Figure 22 shows the angular position and velocity of the solar array during the deployment.

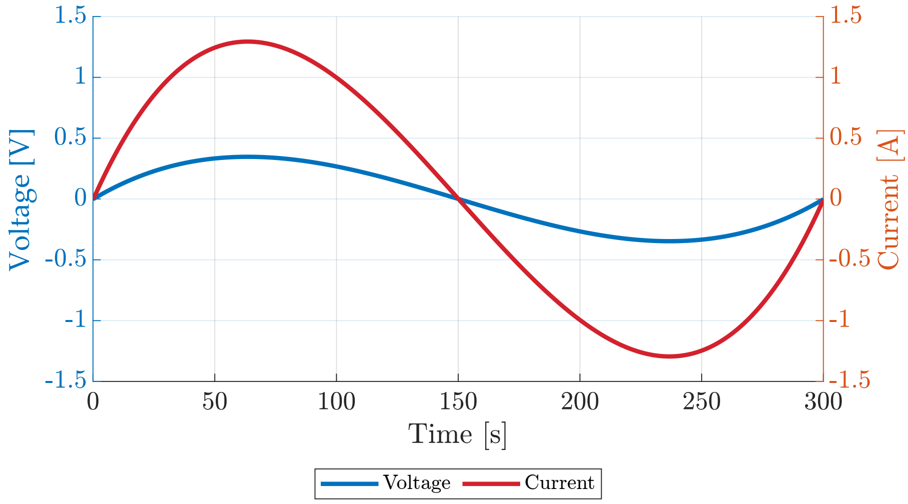

Figure 20: Current and Voltage on the DC motor during deployment

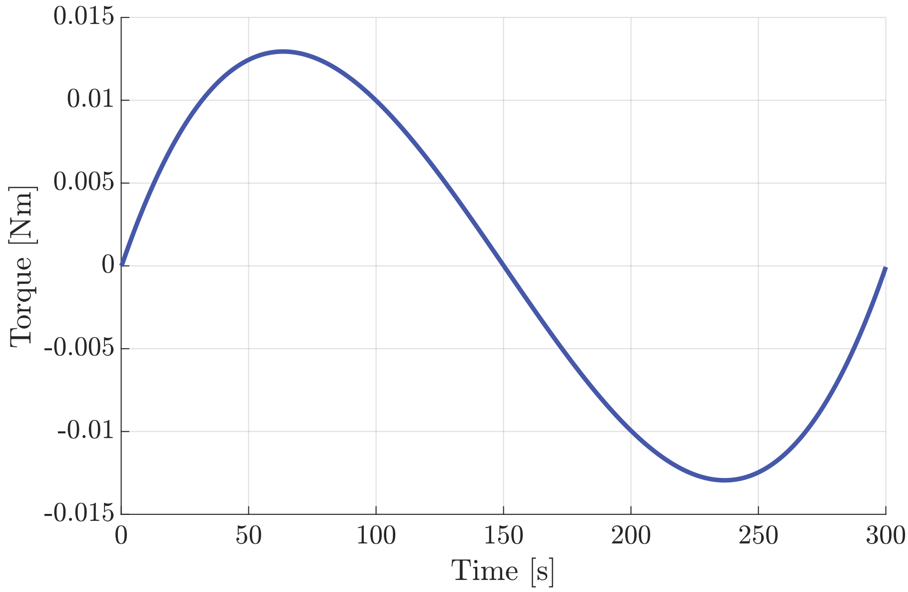

Figure 21: Torque generated by the DC motor during deployment

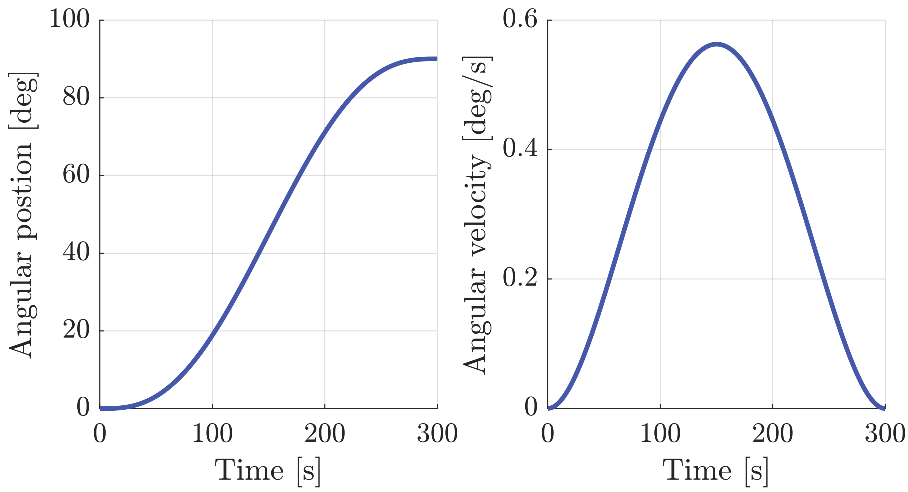

Figure 22: Angular position and velocity of the solar array assembly during deployment

3.3. Parameters generation for MBDyn

3.3.1 Input sequence

The reference position that the motor needs to follow in the MBDyn model has been approximated with a SISO model which best fitted the step5 input. The state-space matrices of the model were obtained using the MATLAB *tfest* function, which computes the transfer function that better fits the provided input-output data.
As it can be seen in Figure 32, the accuracy of the model is quite low for the first few seconds, where a significantly higher amount of torque is required, compared to the other models; after that the computed SISO model is able to follow accurately the step5 input, with some very low-amplitude and low-frequency oscillations once the final position is reached.

3.3.2 Motor parameters

Reproducing in MBDyn the model of the motor and its controller shown in Figure 19 is beyond the scope of this project; instead, the torque required to move the solar array can be generated

by an appropriate spring-damper system.
To find the correct values, the corresponding model of the simplified system has been implemented in simulink. Then the stiffness and damping coefficients have been optimized using the MATLAB built-in function *fmincon* in order to minimize the square of the difference between the torques provided in the two models. The equivalent stiffness and damping coefficient have been found to be $K = 1.722\,e5\ Nm/rad$ and $D = 2.791\,e4\ Nms/rad$ respectively. The results of such optimization are shown in Figure 23. It can be seen that both the torque and the solar array position in the approximated case are almost identical to their corresponding values obtained using the more detailed model.

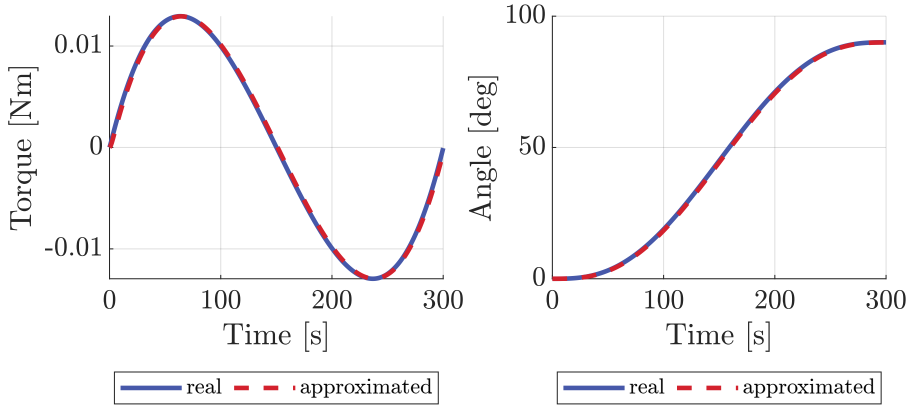

Figure 23: Comparison of torque and angular position between the real system and approximated one

4. MBDyn model

4.1. General Overview

The previously depicted compound, was further simplified to be implemented in MBDyn. In particular the panels were considered as completely rigid and a first analysis was conduced grounding the whole system. In the attempt of sticking to reality as much as possible, the DC motor was modeled through a third order SISO system, while four different springs and two dampers were used to connect the three bodies. Here an initial value problem was solved by means of a multistep integration method, setting tolerances on the derivatives and convergence criteria in the order of 10^{-6}.

4.2. Nodes

Deepening into the model, a series of six nodes was used to individuate the relevant points of the problem. In particular only the three related to the panels were chosen as *dynamic* in the *structural* category, to have inertia properties of the bodies associated to them, while the ones individuating the ground and the actuator were selected as *static*, since no inertia was related to those and therefore saving six degrees of freedom. An additional one individuated as *abstract*, since it only contains the output of the SISO model (used as the imposed motion of the dc motor), was used.

4.3. Reference Frames

A reference frame was attached to each node providing an orientation in space for these. The grounded one was simply oriented as the system of coordinate indicated as *global* in the software, while all the others referring respectively to the $ACTUATOR$ and the the $PANELS$ were rotated for commodity reasons using the Euler angles of -90° around the global *x-axis*. As will be explained in the following paragraphs an extensive use of the *deformable axial joint* will be considered; since one of its characteristic is that of allowing for a rotation only around the *z-axis* of the reference frame in which the joint is defined, a smart definition of these can result as extremely useful, avoiding continuous rotations.

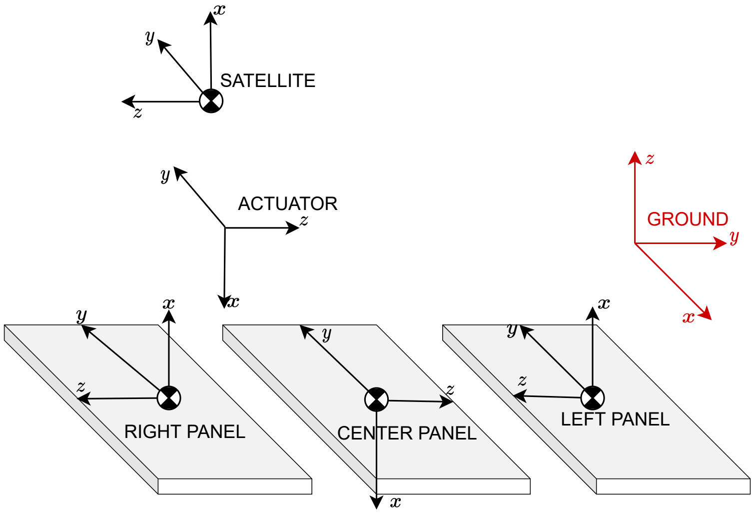

Figure 24: Reference Frames

4.4. Elements

Three *body elements* were attached to the structural dynamic nodes individuating the panels. Mass and inertia properties were therefore specified as illustrated in Appendix A.

As explained in subsection 3.3, the input of the DC motor was reconstructed using a state space SISO of third order which, coupled with a deformable axial joint, drives the position of the the main assembly during the first act of the deployment. The SISO is individuated in the code as *genel* (standing for generic element) whose output is stored by means of a *drivecaller* into the abstract node previously mentioned.

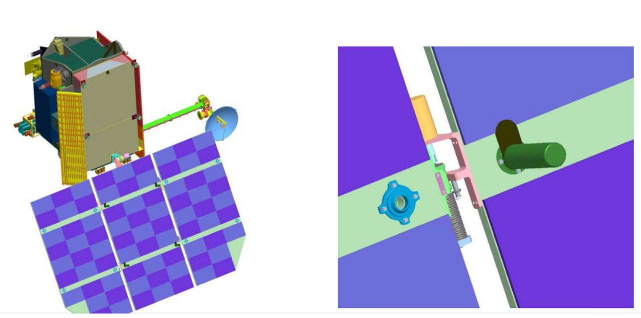

Figure 25: Hinges details

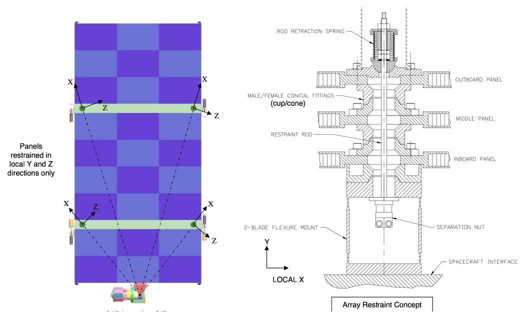

Figure 26: Hinges' details

A set of twelve joints were in the end used in order to correctly constraint all the nodes in a proper way. Starting from *NODE GROUND*, this was fixed through a clamp constraining all its degrees of freedom. Two more linking systems were added so to attach the *ACTUATOR* node to the *GROUND* one. In particular a *total joint* allowing only for the rotation around the *z-axis* in the *ACTUATOR reference frame*, was used. Overlapped to the latter, a *deformable axial joint* was inserted to simulate through an equivalent spring-damper system, prestrained by means of the above mentioned motor input, the rotation of the assembly. The degrees of freedom of the middle panel were fully constrained to the *actuator node* through a full active *total joint*, making these two to act and therefore move as a whole.

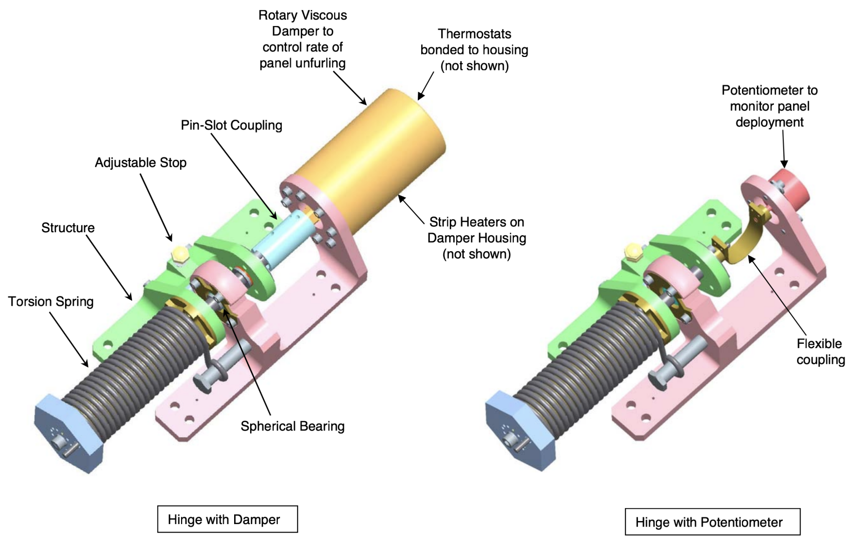

Figure 27: Panel to panel real hinges

Figure 27 shows the real hinges mounted in the side to side panel connections. These consist of a double mechanisms presenting a prestrained spring in both cases to which a damper in one and a potentiometer in the other are attached. In the attempt of sticking with reality as much as possible, this double connection was replicated in the code using four *total joints* overlapped with an equal number of *deformable axial joints* among which two were equipped with dampers. A series of *bistops* governing the activation and deactivation of these joints were used to allow for the correct deployment sequence. In particular the first opening is the left-hand one, therefore a law is used to make this happen only once the whole assembly reached the 90° of deployment. The one on the right side instead starts opening once the left one reached the 60° of aperture. All the springs are prestrained within an angle of 180°, which together with the bistop contributes to correctly arrest the panels in their motion once these are fully deployed.

4.5. Results

As anticipated in the introduction, the goal was to use MBDyn to analyze the movements of the system developed and the torque required, once the motor and the spring-damper mechanism of the panel to panel linkers were sized through Matlab and Adams.
The deployment sequence consisted of two independent movements, differently governed. The first one, guided through the DC motor, is aimed at bringing the whole assembly to a 90° configuration with respect to the satellite body, to which the panels were laid on. As shown on Figure 28, the actuation system is sized to end this first act of the deployment in 300s. Because of a required initial acceleration different from zero, an initial peak in the angular velocity can

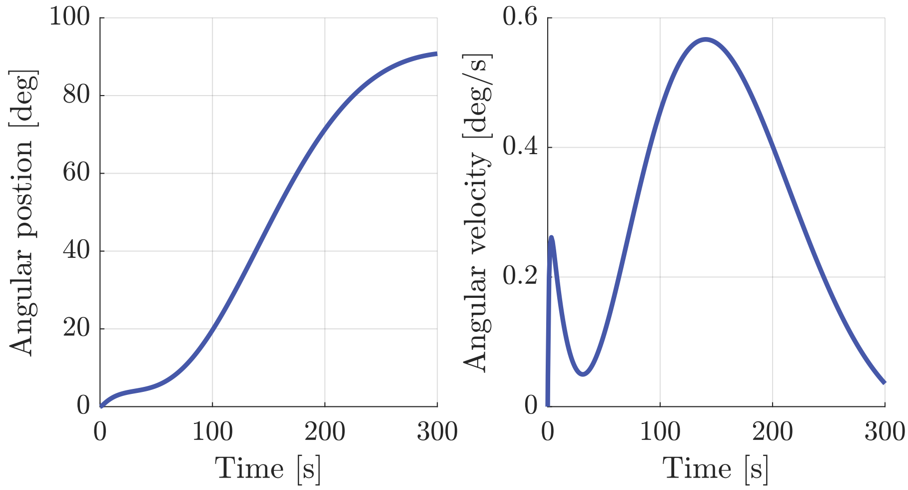

Figure 28: First Deployment

be seen. The second peak is due to a progressive increment of the deployment velocity which is then slowed down once the system starts approaching the final position. Confirmations of previous statements are found in the image displaying the behavior of the angular position where a curvature change is evident in the first 20 seconds of the simulation (impulsive start followed by the deceleration), while a the position tend to stabilize around 90°, due to the angular velocity of the actuator being brought to zero.
For what concerns the lateral panels' deployment, similar considerations can be done. The body initiating the opening sequence is that placed on the left, looking at the system from the satellite body. The input is as impulsive as that given by a precharged spring can be. Therefore a sudden acceleration is experienced, bringing the angular velocity to reach is maximum in the first 20 seconds of the simulation; as a consequence great part of the movement occurs in less that 50 seconds after which a flattening behaviour of the angular position curve, caused by the dampers, is displayed. The right panel is set into motion when the left one reaches 60° in the opening process and shows an identical behaviour in terms of angular position and rate (as expected being the two bodies equal as well as the joints system).

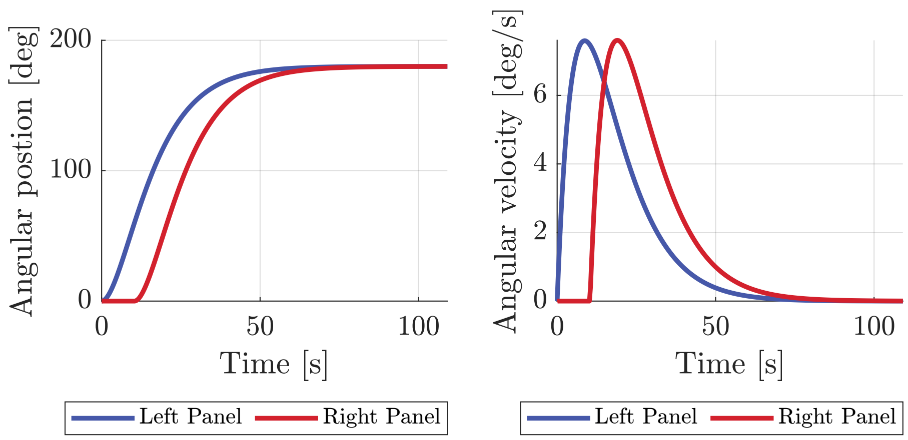

Figure 29: Second Deployment

The behaviour of the reactant torque at the actuator, exactly resembles what was shown in the second graph of Figure 28. The impulsive input of the motor results in a counter couple presenting a negative peak towards the very beginning of the deployment sequence. It is worth noticing that during the two deceleration processes a negative concavity of the curve is shown while a positive one follows the speed increasing mechanism. However being the movement slow and particularly smooth, after the first peaks, just what seems to be small oscillations around the zero are manifested, but they match the couple obtained through the motor sizing.

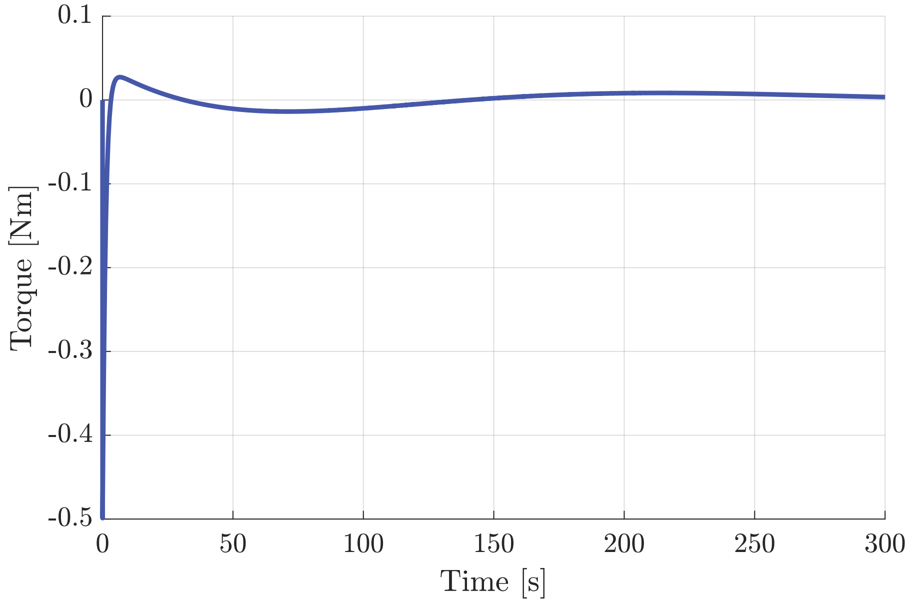

Figure 30: Torque at the actuator

4.6. Satellite

A further development of the previous model included the substitution of the grounded node with a body representing that of the satellite, left free to move in space. No substantial changes were made to the above explained construction, but for fixing the ACTUATOR to the SATELLITE in a way that only the rotation about the *z-axis* was allowed (in the actuator reference frame) as was done in the previous system with respect to the ground node. The SATELLITE itself was considered as a simple *element:body* attached to a *structural dynamic node*, for which inertia properties and mass were specified according to the information found in literature.
Hereafter, Figure 31 displays the rotation angles of the whole assembly (which is solidly moving with the satellite body). These are computed with respect to the global reference frame, neglecting the initial rotation between this coordinate system and that attached to the satellite body. As expected, due to the inertia of the panels, during both their opening motions, some minor displacements are followed as response by the system. In particular, rotating the whole solar array causes a significant rotation of the satellite's main body around the same axis (yellow curve). The imbalance generated by the asymmetric deploying of the side panels (one after the other) is clearly visible also (peaks). However the overall effect of the side panels opening cancels out once they both converge to the 180°, resulting in a negligible rotation around the x and y axes of the ground reference frame.
The movements are not changing the spatial arrangement of the assembly in terms of translations; this happens thanks to the huge mass of the satellite if compared to that of the panels and to the small displacements of the baricenter of the whole system.

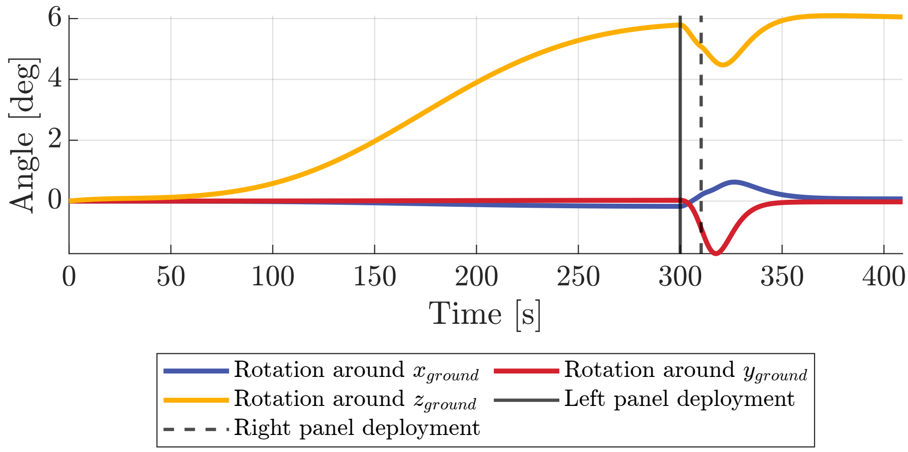

Figure 31: Satellite rotations around the ground reference system

5. Results Comparison

All the results previously presented are here gathered together. As expected, having followed the exact same procedure in Adams and Matlab, very minor discrepancies are shown among the two models, while a significant difference is shown at the beginning of the simulation comparing these two with the MBDyn one. Reason of such behaviour lays in the way the motion of the actuator is driven. The different commanded position causes an impulsive start which reflects in the peaks on the torque and angular velocities.

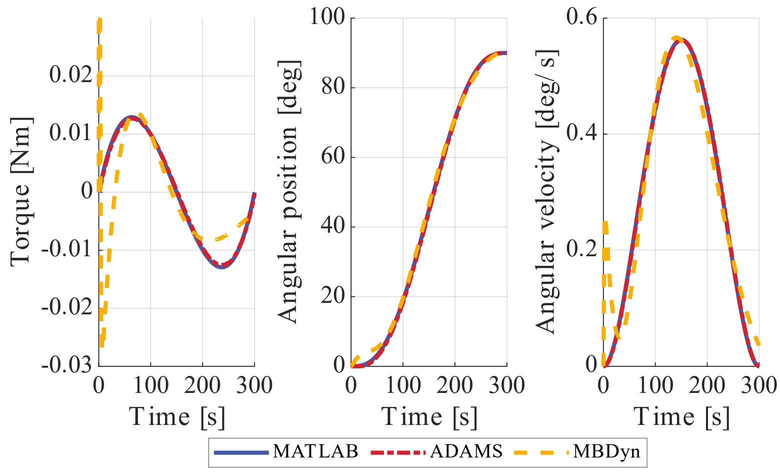

Figure 32: First Deployment

For what concerns instead the lateral deployment, no significant differences are shown. The precharged spring-damper system behaves exactly in the same way in the three models. No dynamic modelling of particular subsystems is here involved therefore there is no reason for the physics to be different in the three cases.

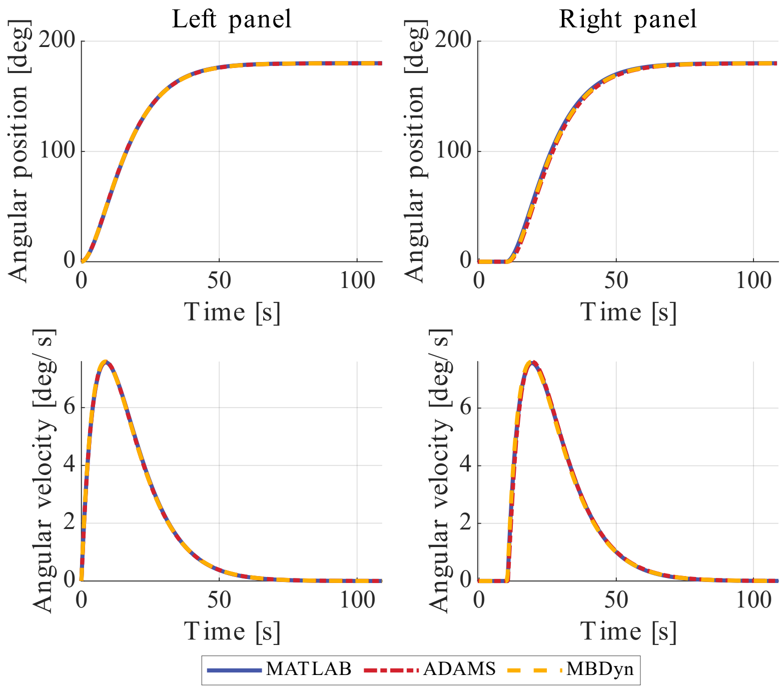

Figure 33: Second Deployment

A. System Data

Satellite			
Satellite Mass	1853	kg	
Satellite Inertia	$\begin{bmatrix} 677.8 & 69.8 & -6.2 \\ 69.8 & 1038.6 & -32.5 \\ -6.2 & -32.5 & 1192.9 \end{bmatrix}$	$kg \cdot m^2$	ref. frame: see Figure 1, Figure 24

Table 1: Satellite properties

Solar Array			
Center Panel Mass	20.208	kg	ref. frames: see Figure 24
Side Panels Mass	14.451	kg	
Center Panel Inertia	$\begin{bmatrix} 16.616 & 0 & 0 \\ 0 & 13.507 & 0 \\ 0 & 0 & 3.118 \end{bmatrix}$	$kg \cdot m^2$	
Side Panels Inertia	$\begin{bmatrix} 11.079 & 0 & 0 \\ 0 & 8.989 & 0 \\ 0 & 0 & 2.093 \end{bmatrix}$	$kg \cdot m^2$	
Panels length	2.8	m	
Panels width	1.347	m	

Table 2: Solar Array properties

Panels Joints		
No. of Joints	2 per side panel	
No. of Springs	2 per side panel	
No. of Dampers	1 per side panel	
Springs Stiffness	1e-3	Nm/rad
Springs pre-strain	180	deg
Damping Coefficient of the Damper	2	Nms/rad

Table 3: Joints properties

DC Motor		
Peak Torque	1.5	Nm
Motor Constant	0.01	Nm/A
Resistance	0.268	Ohm
Inductance	0.076	mH
Inertia	0.064e-5	$kg \cdot m^2$

Table 4: DC motor parameters

Controller gains			
PI		**PD**	
$P + I\frac{1}{s}$		$P + D\frac{N}{1 + N\frac{1}{s}}$	
P	0.239	P	59.81
I	841.9	D	565.55
		N	62.81

Table 5: Controller gains

References

[1] A. Bartels, M. Reden, L. Hartz, J. Baker, S. Scott, and T. Jones, "LRO preliminary design review," Tech. Rep., 2006.

[2] MOOG. Db-1500 matrix motor technical datasheet. [Online]. Available: https://www.moog.com/content/dam/moog/literature/MCG/DB-1500.pdf